

Department of Computer Technology
College of Computer Studies
De La Salle University
CTTHES2 Milestones Form

Thesis Title: Cross-Layer Dataset Design and Exploratory Analysis of ESP32-Based ESP-WIFI-MESH Network

Date: June 6, 2026

Proponents
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*Adviser Endorsement

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Milestone: Firmware Development for All Node Roles		Weight (%): 15%
Description	Panel Remarks	
Build the foundational ESP-WIFI-MESH platform as specified in the THES1 stable, fully-instrumented mesh running on real ESP32 hardware, before any attack code is added. Scope: - Root node firmware: experiment phase controller, probe sink, broadcast of phase IDs. - Victim node firmware: probe generator at a fixed interval, with a periodic cross-layer telemetry sampler. - Common code modules used by every node (written once, reused by root, victim, and later the attackers): - Mesh setup — code that connects the board to the ESP-WIFI-MESH network at startup. - Phase listener — a background task that receives the experiment-phase signal broadcast by the root and tags every logged data row with the current phase (baseline / attack / cooldown). - CSV logger — code that writes the telemetry data to a file on the ESP32's onboard flash storage. - Attacker firmware and multi-topology deployment are intentionally deferred to the second and third milestones. Criteria - All firmware variants compile without warnings. - A 3-node test mesh forms within 60 seconds of root startup, with correct parent-child relationships. - Root broadcasts phase transitions and every node receives and applies them within 1 second. - Each node records cross-layer telemetry (RSSI, layer, parent, packet counters) to local flash at the firmware's configured sampling rate, with no missing samples.		

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• A full 10-minute baseline run completes end-to-end; all CSVs • are retrievable from each ESP32 via USB serial.	
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Milestone: Application-Layer Attack Module Implementation		Weight (%): 15%
Description	Panel Remarks	
The ESP32 Wi-Fi firmware is closed-source binary, so attacks are emulated at the application layer using normal esp_mesh_send / esp_mesh_recv calls (no raw 802.11 frames are touched). Blackhole attack — what it does: • A single attacker node positioned to receive probe traffic from victims. • Victims address probes directly to the attacker's MAC (behavioral equivalent of a false short-route advertisement). ◦ Baseline / cooldown phases: attacker forwards probes normally to root. ◦ Attack phase: attacker silently stops calling esp_mesh_send — packets vanish. • Observable signature: zero forwarded probes reach the root during the attack window. Wormhole attack — what it does: • Two attacker nodes: Attacker B placed near victims, Attacker A placed near root. • A wired UART link between A and B acts as the out-of-band tunnel (this is what makes it a wormhole rather than ordinary forwarding). ◦ Attacker B captures probe metadata from local mesh traffic during the attack phase. ◦ Metadata is shipped through UART to Attacker A; CRC-protected to detect errors. ◦ Attacker A reconstructs a replica probe and re-injects it toward root via the legitimate mesh path. • Observable signature: the same logical probe arrives at root twice — once via slow multi-hop, once via fast wormhole shortcut, with a measurable latency mismatch. Criteria: • Blackhole: root logs show the expected drop in arrivals during the attack window and normal arrivals before and after. • Wormhole: root logs show duplicate probe arrivals with measurable latency difference during the attack window. • Both attacks toggle cleanly on phase transitions; no leakage into baseline windows. • Behavior is consistent across all four topologies		

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Milestone: Multi-Topology Testbed Deployment		Weight (%): 15%
Description	Panel Remarks	
Physically deploy and verify the mesh in all four topology configurations from the proposal. Topologies to deploy: - Star — every node connects directly to root (single-hop tree). - Tree — hierarchical structure with intermediate forwarding nodes.		

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- Linear chain — nodes arranged in a line; each reaches only adjacent neighbors (forces long multi-hop paths). - Partial mesh — some nodes have multiple potential parents (most realistic for real-world deployments). Setup constraints: 5 to 10 ESP32 nodes, positions fixed and documented per topology. Criteria: - Each topology converges to its intended parent-child structure within 60 seconds. - Structure is verified by inspecting parent-MAC and layer values in root logs. - Mesh remains stable through a full 5-minute baseline phase (no spontaneous re-routing). - Both attacks show their expected signatures in all four topologies.	
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Milestone: Phase-Controlled Experiment Execution		Weight (%): 15%
Description	Panel Remarks	
Run the full experimental matrix and produce labeled raw telemetry per run. Run timeline (per experiment): 1 minute — mesh formation. 5 minutes — baseline (normal operation). 3 minutes — attack phase active. 2 minutes — cooldown. Experimental matrix: 4 topologies × 2 attack types × at least 3 repeats = minimum 24 runs. - Root broadcasts phase transition signals; every node embeds the current phase ID in its log rows (the ground-truth label). Criteria: - At least 24 complete runs are collected. - Every run's per-node CSV logs are intact. Phase IDs in node logs match the root's broadcast timeline within tolerance.		

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Milestone: Raw Data Extraction and Integrity Validation		Weight (%): 10%
Description	Panel Remarks	
After each run, retrieve and validate the per-node CSV files before they enter the analysis pipeline. Extraction: - Pull CSV files from each ESP32 via USB serial after each run. - Store with run metadata (topology, attack type, repeat number, collection date). - Validation script checks: - Sample coverage — at least 95% of expected samples per node at the configured telemetry rate. - Phase labels populated for every log row. - No file corruption or premature truncation.		

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Criteria: Validation report confirms at least 24 clean runs in the final dataset. Any run failing validation is flagged for repeat collection.	
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Milestone: Cross-Layer Data Preprocessing Pipeline		Weight (%): 10%
Description	Panel Remarks	
Convert raw per-node CSVs into a clean, analyzable windowed dataset. Pipeline steps: - Per-node timestamp re-basing (each node's clock starts at 0). - Cumulative-counter to per-window delta conversion. 5-second non-overlapping window aggregation. - Discard windows with too few valid samples relative to the expected count per window. - Modal phase-label assignment per window (the ground-truth label for downstream analysis). Criteria: - Pipeline accepts a folder of raw CSVs and outputs one unified windowed DataFrame. - Window label distribution matches the experiment schedule within 2% tolerance. Pipeline is deterministic — same inputs always produce the same output.		

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Milestone: Cross-Layer Feature Engineering (16 Features)		Weight (%): 10%
Description	Panel Remarks	
From each windowed sample, compute the 16 cross-layer features defined in Table 4.11 of the proposal. Feature specifications may iterate as the EDA reveals which carry useful signal. Feature families: - Forwarding behavior — Forwarding Ratio, Ingress-Egress Delta, Consistency Score. - Link reliability — Retry Rate, Packet Delivery Ratio (PDR). - Topology stability — Parent Switch Rate, Layer Change Count, Hop Stability Duration. - Physical layer — RSSI mean, RSSI variance, RSSI stability. - Cross-layer inconsistencies — RSSI vs Hop Count Difference, Latency vs Hop Count Ratio. - Auxiliary tunnel-derived features (3). Implementation note: - PDR and Latency-Hop features require joining each node's outgoing log with the root's probe-arrival log on (src_node_id, sequence_number). Criteria: - Feature table contains all 16 columns with non-null values. - No feature is uniformly NaN.		

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Milestone: Exploratory Data Analysis		Weight (%): 10%
Description	Panel Remarks	

Explore the feature dataset for visual and statistical evidence of attack separability. Analyses produced: • Descriptive statistics per feature (mean, std, range, quartiles). • Distribution plots split by attack label (baseline vs blackhole vs wormhole). • Spearman correlation heatmap across all features. • 2D dimensionality reduction — PCA and t-SNE, points colored by ground-truth label. Goal: • Identify which features carry discriminative signal. • Confirm visually whether baseline and attack windows are separable. Criteria: • EDA report identifies at least one feature whose distribution clearly differs between baseline and each attack type. Quantitative backup is provided (effect size, distribution overlap, or similar measure).	
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Milestone: Unsupervised Clustering Suite		Weight(%): CTTHES3
Description	Panel Remarks	
Apply unsupervised clustering algorithms to the cross-layer feature dataset. The primary and baseline algorithms must complete in CTTHES2; the three comparative algorithms can be moved to CTTHES3 if time becomes tight. Algorithms (target: all 5 in CTTHES2): • DBSCAN (primary) — eps selected via k-distance plot. • K-Means (baseline) — k selected via elbow and silhouette analysis. • Hierarchical clustering (comparative). • Gaussian Mixture Model (comparative). • Spectral clustering (comparative). Constraints: • All algorithms see identical feature input for fair comparison. • No labels are used during clustering — labels only enter the evaluation Criteria (CTTHES2 minimum): • DBSCAN and K-Means run end-to-end and produce a cluster label for every window. • Hyperparameter choices are documented and justified. Comparative algorithms ideally finished in CTTHES2; remaining work, if any, carries over to CTTHES3 Sensitivity and Robustness Analysis.		

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Milestone: Cluster Evaluation and Characterization		Weight(%): CTTHES3
Description	Panel Remarks	
Evaluate clustering quality and interpret what each cluster represents. Internal validation (no labels used): • Silhouette coefficient. • Davies-Bouldin index. Post-hoc validation (against ground truth): • Cluster purity. • Adjusted Rand Index (ARI). Cluster characterization: • Centroid heatmap showing per-feature z-scores for each cluster.		

- Mapping of clusters to {baseline, blackhole, wormhole} groupings with quantitative confidence. Criteria: - A 5-algorithm $\times$ 4-metric comparison table is produced. The best-performing algorithm's clusters are mapped to the three ground-truth classes.	
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Milestone: Sensitivity and Robustness Analysis		Weight(%): CTTTHES3
Description	Panel Remarks	
Test how sensitive the results are to the design choices, and absorb any carry-over from CTTTHES2 clustering work. Sensitivity dimensions: - Window length (3s, 5s, 10s). - DBSCAN parameter ranges (eps, min_samples). - Feature-subset selection (which features contribute the most signal). Robustness checks: - Cluster-assignment stability across additional experimental runs. Completion of any comparative clustering algorithms.		

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Milestone: Extended Experimental Runs		Weight (%): CTTTHES3
Description	Panel Remarks	
Strengthen statistical reliability with additional repeats and capture more environmental variation. - Target at least 5 repeats per topology-and-attack combination. - Capture variation across time-of-day to include differing Wi-Fi interference patterns. - Re-run any topology that showed unstable convergence in CTTTHES2.		

Instructions:

- Every milestone in your research should have a corresponding milestones table.** Milestones can be a module, process, output, document, algorithm, test, phase, and the like. Replicate the tables above as many times as you need. For each table, ensure that the total weight of all milestones would be 100% for your CTTTHES2. Note this would be used by your Thesis Panel as a reference during your CTTTHES2 demo, to determine completion.
- Note to **indicate all modules needed for the completion of the entire thesis** (both CTTTHES2 and CTTTHES3), however indicate **CTTTHES3** in the **Weight** for modules that would be implemented on CTTTHES3. Ideally the milestone to be completed during your CTTTHES2 stage would be the bulk of your thesis work, leaving CTTTHES with tasks such as; final thesis document writing, implementation of additional tests, fine tuning, completion of error handling, and other finalization tasks.

----- DO NOT REMOVE - FOR CTTHES2 DEMO USE ONLY -----

Proponents	
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Final Grade: (70% <i>passing mark</i>)	
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Recommendations for THES3:

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